AI 101 for Mechanical Engineers · Module 7

Fit a line and interpret coefficients

Lesson 7.1

What you will learn

> Interpret a linear regression equation, identify coefficient units, and calculate a prediction and an observation-minus-prediction residual.

Lesson overview

Module 7 · Lesson 1

Fit a line and interpret coefficients

AI 101 for Mechanical Engineers

1 Read the model and its units

$$\Delta T = b_0 + b_1 P$$

Predicts temperature rise from heater power.

Symbol	Meaning	Unit
ΔT	Temperature rise	K
P	Heater power	W
b_0	Intercept: predicted rise at zero power	K
b_1	Slope: change in rise per watt	K/W

K/W × W = K

2 Calculate a prediction and residual

Given (fitted coefficients and input)

b_0 = 1 K | b_1 = 0.6 K/W | P = 10 W

Predicted rise

1 K + (0.6 K/W × 10 W) = **7 K**

Measured rise: 8 K

$$r = y - \hat{y} = 8\ \text{K} - 7\ \text{K} = +1\ \text{K}$$

y: measured rise · $\hat{y}$: predicted rise · r: residual

A positive residual means the model underpredicts this observation.

3 Separate fitting from prediction

Least squares (fitting)

Chooses coefficients that minimize the sum of squared residuals.

Inference (prediction)

Uses the fitted equation to calculate a prediction for a new input.

"Linear" means linear in the coefficients. A fitted polynomial can still be linear in this sense.

A fitted coefficient is not automatically a physical property or proof of a cause. Interpret it using the measurement range and model assumptions.

Key takeaway

Keep coefficient units and residual signs clear.

Moving the line changes the model; it does not create new measurements.

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Linear regression models a numerical response

$$\Delta T = b_0 + b_1 P$$

- Uses fitted coefficients [16]
- ΔT: temperature rise in K; P: heater power in W
- b_0: intercept in K; b_1: slope in K/W

Read each coefficient with its unit

- Intercept: predicted rise at zero input power
- Fitted b_0 = 1 K; 0 W may lie outside measured range
- Slope: change in predicted rise per unit change in power
- K/W × W gives K, the output unit

> A fitted coefficient is not automatically a verified physical property.

Least squares fits; the equation then predicts

Fitting

- Minimizes the sum of squared residuals [16]

Applying

- Uses the fitted equation on a new input

> “Linear” means linear in the coefficients: a polynomial fit can be linear even though its curve bends.

The predicted rise at 10 W is 7 K

$$\Delta T_predicted = 1\ \mathrm{K} + (0.6\ \mathrm{K/W} \times 10\ \mathrm{W}) = 7\ \mathrm{K}$$

- b_0 = 1 K and b_1 = 0.6 K/W
- The measured rise is 8 K

A residual of +1 K means underprediction

$$r = y - \hat{y} = 8\ \mathrm{K} - 7\ \mathrm{K} = +1\ \mathrm{K}$$

- y: measured rise; ŷ: prediction; r: residual; all in K
- Positive sign: the prediction is below the measurement

Adjusting a line is not new evidence

- Adjusting the line changes the proposed model
- It collects no new measurement and performs no experiment
- The calculation is illustrative, not a validated industrial recommendation

Several inputs extend the same idea

$$y_{hat} = b_0 + b_1 x_1 + b_2 x_2$$

- Each coefficient: change per unit of its feature, others fixed
- Units: output units divided by that feature’s units
- The fitted association is not automatically causal or physical

CHECK YOUR UNDERSTANDING

What units must the slope have when predicting temperature rise in K from heater power in W?

Answer before moving on.

CHECK YOUR UNDERSTANDING

What units must the slope have when predicting temperature rise in K from heater power in W?

> K/W. Multiplying by the input power in W produces a temperature-rise term in K.

CHECK YOUR UNDERSTANDING

What does the residual +1 K mean in the supplied example?

Answer before moving on.

CHECK YOUR UNDERSTANDING

What does the residual +1 K mean in the supplied example?

> The measured rise exceeds the prediction by 1 K, so the model underpredicts that observation.

Key takeaways

- Interpret coefficients together with their units and assumptions
- Least squares fits coefficients; inference applies the fitted equation
- Residual = measured output minus predicted output in this course
- Adjusting a model does not create new experimental evidence

References

[16] G. James, D. Witten, T. Hastie, and R. Tibshirani, An Introduction to Statistical Learning: With Applications in R, 2nd ed. New York, NY, USA: Springer, 2021. doi: 10.1007/978-1-0716-1418-1

Numbers match the course reference list. The written lesson is the authoritative text.